Aswini Kumar Padhi

Microsoft Research India PhD Fellow

: Google Scholar, : Linked In, : GitHub, : Personal Website

Email : aswinikumarpadhi1995@gmail.com
eez238359@iitd.ac.in
Mobile : +91-7978014958

Education

- **Indian Institute of Technology, Delhi** — New Delhi, India
  Doctor of Philosophy *in* ***Natural Language Processing****; GPA: 9.00/10* — *Jul 2023 - Present*
  - ***PhD Thesis:*** *"Leveraging Computational Methods to Combat Hate Speech via Multi-faceted Automated Refutation Systems Generation on Social Media",* ***Supervisor:*** *Prof. Tanmoy Chakraborty*
  - ***Courses Credited:*** *Social Network Analysis (ELL880), Data Visualization (ELL824), Deep Learning for Natural Language Processing (ELL884), Cloud Computing (ELL887)*
- **ABV-Indian Institute of Information Technology, Gwalior** — Gwalior, India
  Master of Technology *in* ***Computer Science and Engineering****; GPA: 9.34/10* — *Aug 2021 - May 2023*
  - ***M.Tech Thesis:*** *"I2C SLR: Indian Continuous Sign Language Recognition System",* ***Supervisor:*** *Dr. Sunil Kumar*
- **Veer Surendra Sai University of Technology, Burla** — Burla, India
  Bachelor of Technology *in* ***Computer Science and Engineering****; GPA: 8.4/10* — *Aug 2012 - May 2016*
  - ***B.Tech Thesis:*** *"Efficient FPGA and ASIC Realization of FIR Digital Filters Using Distributed Arithmetic",* ***Supervisor:*** *Prof. Amiya Kumar Rath*

Research Interests

- Large Language Model (LLM) Safety, Alignment, and Robustness
- Reinforced Language Modeling
- Diffusion-based Language Modeling
- Online Safety

Honors and Awards

- Selected for **Microsoft Research India Academic Summit, 2026.**
- Presented our work at **Microsoft Research India Lab, 2026.**
- Selected as **Institutional Research Associate (IRA)** at IIT Delhi-AbuDhabi campus for Department of Computer Science and Engineering for the period: *5th February-2026 to 31st May-2026.*
- Invited to present ACL 2025 paper **"Counterspeech the Ultimate Shield! Multi-Conditioned Counterspeech Generation through Attributed Prefix Learning"** at **ACM ARCS, India 2026.**
- Selected as Visiting Researcher at the University of Minnesota (January 2026), under the `NSF-DST` project, Travel period: *1st January-2026 to 31st January-2026.*
- Delivered talk on **"Agentic Safety"** at **Microsoft Research India Lab, 2025.**
- Joined as Research Intern at Microsoft Research India Lab, Bangalore, India, *September-2025 to November-2025.*
- Received **Microsoft Research India and ACM** travel grants for attending ACL 2025.
- Our proposal (Me and Prof. Tanmoy Chakraborty) on, **"Diffusion Language Models"** has been selected as a finalist for the **Qualcomm Innovation Fellowship 2025 India**, chosen from over 150 proposal submissions.
- Received **Microsoft Research India PhD Fellowship 2025** by Microsoft Research India (MSRI).
- Selected for **Microsoft Research India Academic Summit, 2025.**
- Delivered talk on **"Large Language Model Safety and Alignment"** at **ABV Indian Institute of Information Technology, Gwalior.**
- Selected for **Microsoft Research India travel grant** to travel to Mexico City for NAACL 2024.
- **Ranked** 1^{st} in the Department of Computer Science and Engineering, IIITM (Gwalior), among the post-graduating batch of 2023.
- Qualified among the top 2% of the students (about 160,000) appearing for *Graduate Aptitude Test in Engineering (GATE)*, 2021.
- Received National Cadet Corps (NCC) 'C' certificate during *NCC Examination*, 2010.

Publications

***: Equal contribution**

Journals

[J1] **Aswini Padhi**, Ridhankur Sarkar, Nilesh Ranjan Pal and Tanmoy Chakraborty. Counter in Turns! Automated Refutation Systems Generation in Multi-Turn Dialogue Using Reward-informed Learning (Under Review)

[J2] **Aswini Padhi**, Apoorv Jain, and Tanmoy Chakraborty. Counter with peace! Persona Guided Time-Dependent Discrete Stochastic Processes for Conditioned Automated Refutation Systems Generation (Under Review)

Conferences

[C1] **Aswini Padhi**, Mrinal Devnath, and Tanmoy Chakraborty. Automated Refutation Systems, the power in words! Optimizing Student Model towards Rationale-Aligned Automated Refutation Systems Generation (Under Review)

[C2] Sujoy Nath, **Aswini Padhi**, and Tanmoy Chakraborty. Counter with Evidence! A Multi-Agent Memory Efficient Reasoning Framework for Hate Category Informed Automated Refutation Systems Generation (Under Review)

[C3] **Aswini Padhi**, Anil Bandhakavi, and Tanmoy Chakraborty. 2025. Automated Refutation Systems the ultimate shield! Multi-Conditioned Automated Refutation Systems Generation through Attributed Prefix Learning. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 27643–27663, Vienna, Austria. Association for Computational Linguistics.

[C4] Amey Hengle*, **Aswini Padhi***, Anil Bandhakavi, and Tanmoy Chakraborty. 2025. CSEval: Towards Automated, Multi-Dimensional, and Reference-Free Automated Refutation Systems Evaluation using Auto-Calibrated LLMs. In Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), pages 5402–5419, Albuquerque, New Mexico. Association for Computational Linguistics.

[C5] Amey Hengle*, **Aswini Padhi***, Sahajpreet Singh, Anil Bandhakavi, Md Shad Akhtar, and Tanmoy Chakraborty. 2024. Intent-conditioned and Non-toxic Automated Refutation Systems Generation using Multi-Task Instruction Tuning with RLAIF. In Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), pages 6716–6733, Mexico City, Mexico. Association for Computational Linguistics.

[C6] S. Rajpopat, **A. K. Padhi** and S. Kumar, "An Efficient Correlated Gamma Wave-Based Lightweight Model for Schizophrenia Detection," 2023 IEEE International Conference on Computer Vision and Machine Intelligence (CVMI), Gwalior, India, 2023, pp. 1-6, doi: 10.1109/CVMI59935.2023.10465207.

Services

- Teaching Assistant for *Computer Architecture* course - 2026-27 at *IIT Delhi Abu Dhabi*
- Teaching Assistant for *CEP-GenAI* course - 2026-27 at *IIT Delhi*
- Teaching Assistant for *Advances of LLM* course - 2025-26 at *IIT Delhi*
- Teaching Assistant for *CEP-GenAI* course - 2025-26 at *IIT Delhi*
- Teaching Assistant for *ELL884 (Deep Learning for NLP)* course in Sem-4 of 2025-26 at *IIT Delhi*
- Teaching Assistant for *ELL409 (Machine Learning)* course in Sem-3 of 2024-25 at *IIT Delhi*
- Teaching Assistant for *ELL101 (Introduction to Electrical Engineering)* course in Sem-I of 2023-24 at *IIT Delhi*

Work Experience

- **Wipro Technologies** — Bengaluru, India
  Associate Consultant on Machine Learning *for* ***Client: Chevron USA*** — *December 2016 - June 2019*
  - ***Job Role: Machine Learning-Based Predictive Maintenance System for Oil Equipment:*** *Remotely led the development of a machine learning model for Chevron's oil equipment monitoring from the offshore delivery center in India. Collaborated virtually with on-site engineers to understand the data collected from IoT sensors on approximately 1,000 pieces of rotating equipment. Developed and fine-tuned a random forest algorithm using Python and scikit-learn to predict potential equipment failures based on historical sensor data. The model analyzed parameters such as vibration patterns, temperature fluctuations, and pressure readings to identify early signs of malfunction.*

- **Wipro Technologies** Hyderabad, India
  Associate Consultant on Machine Learning *for* ***Client: VALE S.A.*** *July 2019 - December 2020*
  - ***Job Role: AI-Powered Inventory Optimization for Iron Ore Supply Chain***: *Managed the creation of an AI-powered inventory optimization system for VALE SA's iron ore supply chain from the offshore center in India. Worked remotely with the client's logistics team to gather historical inventory and shipment data. Developed a deep learning model using TensorFlow to forecast iron ore demand across different global markets. The model incorporated external factors such as market prices, seasonal trends, and macroeconomic indicators to improve prediction accuracy. Implemented the model within VALE's existing ERP system through API integrations, providing real-time inventory recommendations.*